



Week 1: What is Encryption?

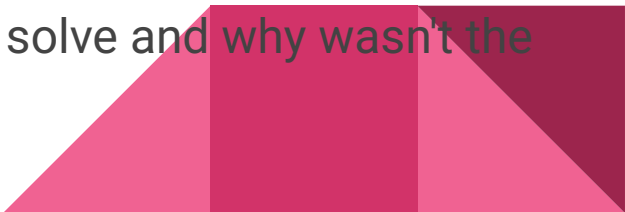
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Overall Objective

How does a client trust who they're exchanging messages with over the internet?



Agenda

- Week 1: What is encryption? How does encryption keep data safe on the internet?
 - Week 2: What do websites use to keep data safe? How does the TLS handshake keep messages safe between the client and server? How does it guarantee future session safety if a previous session is compromised?
 - Week 3: How does a client ensure that the server is who they say they are? How does a server get a certificate signed to verify its authenticity? What are certificate authorities role in keeping the TLS handshake safe for both parties?
 - Week 4: What problem does certificate transparency solve and why wasn't the CA system enough on its own?
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Week 1 Objectives

- Define Encryption and the different types
- Describe the advantages and disadvantages of each type
- Explain the applications of encryption



Encryption is the act of scrambling messages (plaintext) with a key into indecipherable ciphertext and decrypting that ciphertext back into plaintext.

Without encryption, checking your bank account, sending a message, or making a video call would be exposed to anyone willing to intercept them.



Symmetric Encryption

- Symmetric Encryption uses a shared key to encrypt messages (plaintext) into ciphertext and decrypt back into plaintext
- Passed physically between two parties
- The internet made physical key exchange impossible at scale
- Required a new encryption scheme with the advent of the internet



Public Key Cryptography



Hybrid Approach - RSA (Deprecated in TLS 1.3)



Hybrid Approach - Diffie-Hellman (Forward Secrecy)

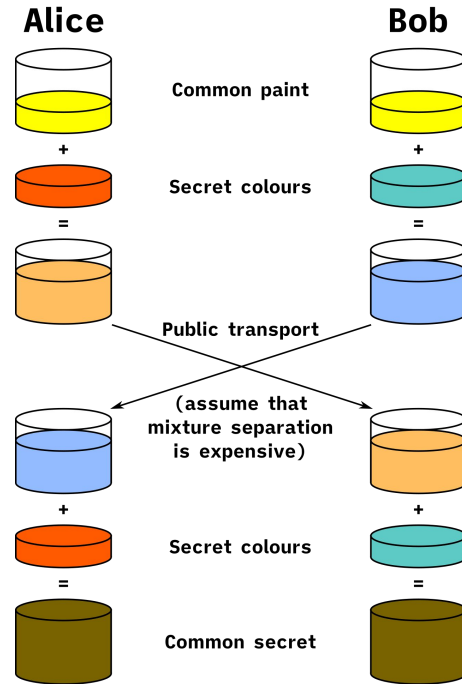


Figure 1: Diffie-Hellman Key Exchange. [Reference](#)

Applications

- Web Browsing (HTTPS, TLS)
- Secure communications (WhatsApp, Signal)
- Digital Signatures (Certificates)
- Authentication (Banking Account, Login Screens)
- Blockchain technology (Crypto)



Next week we'll learn how the browser uses all of this encryption every time you visit a website.



References

https://en.wikipedia.org/wiki/Diffie%E2%80%93Hellman_key_exchange

https://en.wikipedia.org/wiki/Primitive_root_modulo_n

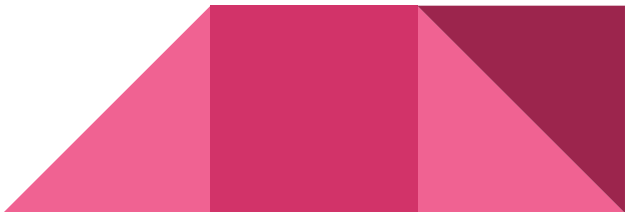
<https://www.geeksforgeeks.org/computer-networks/symmetric-key-cryptography/>

<https://www.cloudflare.com/learning/ssl/how-does-public-key-encryption-work/>

<https://www.cloudflare.com/learning/ssl/what-is-a-session-key/>

https://en.wikipedia.org/wiki/Transport_Layer_Security

<https://www.ibm.com/think/topics/asymmetric-encryption>





Questions?